

1. **Experiment:** Construction of the gp130-caspase9 Suicide Selection System in Raji Cells
2. **Time:** 2026.03.06-2026.03.18
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4. **Materials:** Hek293T cells, Raji cells, 5% CO₂ incubator, Hieff Trans® Polyethylenimine Linear (PEI), pCDH-gp130-caspase9 transfer plasmid (gp130-caspase9 fusion gene synthesized and provided by the company), pMD2.G envelope plasmid, pCMVR8.74 packaging plasmid, DMEM high glucose, RPMI-1640 medium, 10% FBS, 1% penicillin-streptomycin, Polybrene, anti-human gp130 (CD130) antibody (APC or PE conjugated), FACS buffer (PBS containing 2% FBS), 0.45 µm syringe filters, 100 mm culture dishes, 6-well plates, 15 mL conical tubes, centrifuge (Eppendorf 5810 R), flow cytometer (FACS Aria III), biosafety cabinet, and automated cell counter.

5. Method:

- (1) One day before transfection, Hek293T cells were passaged and evenly plated on 100 mm culture dishes (at 9×10^6 cells per dish). Cells were cultured at 37 °C in a 5% CO₂ incubator. The medium was replaced with 10.5 mL of pre-warmed, antibiotic-free medium (5% FBS) 3 h before transfection.
- (2) A mixture of three plasmids (transfer plasmid + envelope plasmid + packaging plasmid) in equimolar amounts was prepared, and transfection was performed using the PEI transfection method with the Hieff Trans® Polyethylenimine Linear (PEI). The transfection mixture composition per well was as follows:

Table 1. Transfection mixture composition.

Transfection tube	Reagent	Volume/Amount
A	PEI (1 mg/mL)	90 µL
	Antibiotic-free medium	910 µL (to 1 mL total)
	Transfer plasmid (pCDH-gp130-caspase9)	15 µg
B	Envelope plasmid (pMD2.G)	5 µg
	Packaging plasmid (pCMVR8.74)	10 µg
	Antibiotic-free medium	Up to 1mL

- (3) Flow cytometric antibody titration: Raji cells were harvested and resuspended in FACS buffer (PBS containing 2% FBS) at 1×10^6 cells/mL. Anti-human gp130 (CD130) antibody conjugated with APC or PE was titrated at graded concentrations of 0.1, 0.25, 0.5, 1.0, 2.5, 5.0, and 10.0 µg/mL in 100 µL per tube. Cells were incubated with the antibody at 4 °C for 30 min in the dark, washed twice with FACS buffer, and analyzed by flow cytometry. The optimal antibody concentration yielding the highest signal-to-noise ratio was determined for subsequent sorting experiments.
- (4) The PEI solution (Tube A) and the plasmid mixture (Tube B) were gently mixed separately, then combined by adding Tube B dropwise into Tube A using a micropipette. The mixture was vortexed briefly and incubated at room temperature for 15 min to allow complex formation. The transfection complexes were then added dropwise to the Hek293T cells and the plates were returned to the incubator.
- (5) Viral supernatant collection: At 48 h and 72 h post-transfection, collect the viral supernatant from Hek293T cells, filter through a 0.45 µm filter, and use immediately or store at -80 °C.
- (6) Cell Inoculation:
 - ① Collect Raji cell from a culture flask.
 - ② Centrifuge at $300 \times g$ for 5 min at room temperature.
 - ③ Carefully remove the supernatant and resuspend the cell pellet in an appropriate volume of complete medium (RPMI-1640 with 10% FBS and 1% penicillin-streptomycin).

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- ④ The cell density was determined using an automated cell counter, and the suspension was diluted to a final concentration of 1×10^6 cells/mL.
 - ⑤ Add the calculated volume of cells to complete medium to achieve a final concentration of 1×10^6 cells/mL.
 - ⑥ For each well of a 6-well plate (or equivalent dish), 2 mL of this cell suspension (2×10^6 cells) was dispensed. If using a 10 cm dish, adjust the volume to 10 mL (1×10^7 cells). The plates were incubated at 37 °C in a 5% CO₂ incubator for 24 h prior to infection.
 - (7) Infection: To each well, Polybrene was added at a final concentration of 8 µg/mL. The multiplicity of infection (MOI) was set to 30, and the specific volume of the virus, calculated as $(30 \times \text{cell number}) / \text{virus titer}$, was added to the wells. The plates were then incubated at 37 °C for 14 h.
 - (8) Flow cytometric sorting: After 60 h of infection, cells were collected by centrifugation at $300 \times g$ for 5 min, and the supernatant containing virus and Polybrene was discarded. Cells were washed twice with PBS and resuspended in FACS buffer at 1×10^7 cells/mL. Cells were washed and stained with fluorescent anti-human Fc and anti-gp130 antibodies at 4 °C in the dark. Cells were washed twice with FACS buffer and resuspended at 5×10^6 cells/mL for sorting. gp130-positive cells were sorted using a FACS sorter (e.g., FACSAria III) and collected into complete medium containing 50% FBS. Sorted cells were cultured at 37 °C, 5% CO₂ for expansion.
 - (9) Expansion and validation of gp130-caspase9-transduced cells: Sorted cells were expanded in RPMI complete medium for 10 days. Transduction efficiency and gp130 surface expression were validated by flow cytometry using the same anti-gp130 antibody. Aliquots were cryopreserved in liquid nitrogen for downstream experiments.